Hopfield Network Exercise

Neural Computing

March 4, 2026

1 Pattern Storage & Weight Matrix Calculation

Consider a Hopfield network with 4 neurons arranged in a 2×2 grid. We will use bipolar representation where each neuron can either be +1 or -1.

We want to store the following two patterns:

- Pattern A: A horizontal line
  $$\begin{bmatrix} +1 & +1 \\ -1 & -1 \end{bmatrix}$$
- Pattern B: A vertical line
  $$\begin{bmatrix} +1 & -1 \\ +1 & -1 \end{bmatrix}$$

Questions:

1. Convert patterns A and B into vectors $\vec{\xi^A}$ and $\vec{\xi^B}$, where each vector has 4 elements.
2. Using the Hebbian learning rule, calculate the weight matrix W. Recall that:
   - $w_{ij} = \frac{1}{N} \sum (\xi_i^\mu \xi_j^\mu) \quad \text{for all patterns } \mu.$
   - The diagonal elements w_{ii} should be set to 0 (no self connections).

2 Pattern Retrieval

3. Consider a new input pattern C

   $$\begin{bmatrix} +1 & +1 \\ -1 & +1 \end{bmatrix}$$

   This pattern has one neuron different from Pattern A. Now:

   (a) Convert pattern C to vector form $\vec{\xi}$.

   (b) Calculate the state of each neuron after one synchronous update. Recall that the update rule is:

   $$O_i(t+1) = sgn\left(\sum_{j}^{N} w_{ij} O_j(t)\right)$$

   (c) Does the network converge to one of the stored patterns after this update? If not, continue the updates until convergence or until you can explain why it won't converge.

3 Energy Function

The energy function of a Hopfield network is given by:

$$E(\vec{x}) = -\frac{1}{2}\sum_{i}^{N}\sum_{j}^{N} w_{ij}\xi_i\xi_j$$

Questions:

4. Calculate the energy of Pattern A ($\vec{\xi^A}$), Pattern B ($\vec{\xi^B}$), Pattern C ($\vec{\xi^C}$).
5. Based on your energy calculations, explain:
   (a) Why do the stored patterns have lower energy values?
   (b) How the network uses energy minimization to perform pattern completion?